

Defects in Crystals

INTRODUCTION

- Crystalline material has a crystal structure in which the atoms are positioned in perfect ordered pattern which is repetitive over large atomic distance.
- Defects have a profound impact on the macroscopic properties of materials.
- Sometimes defects are deliberately created to improve properties of crystal.

CLASSIFICATION OF DEFECTS BASED ON DIMENSIONALITY

0D (Point defects)

Vacancy

Interstitial

Frenkel
defect

Schottky
defect

1D (Line defects)

Edge
Dislocation

Screw
Dislocation

2D (Surface / Interface)

Grain
boundary

Twin
boundary

Stacking
faults

3D (Volume defects)

Dispersant

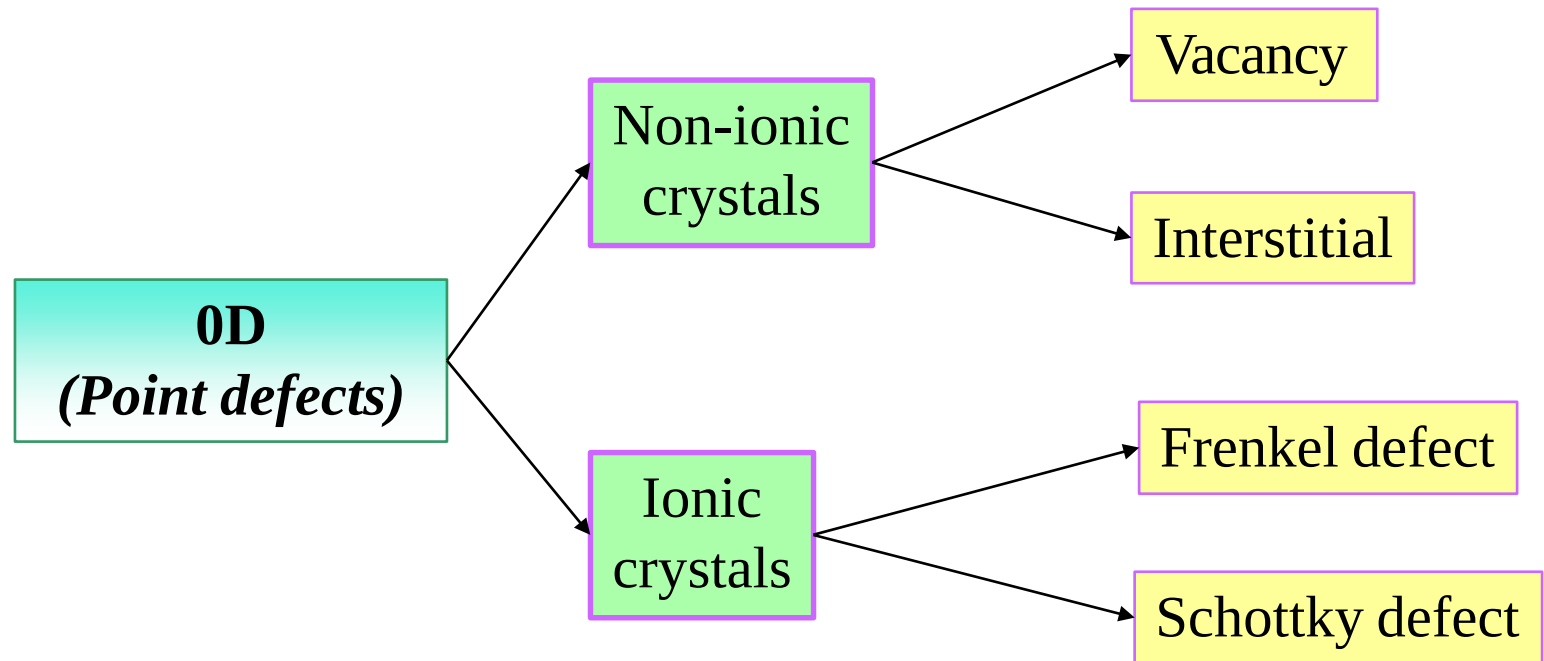
Precipitate

Inclusion

Voids /
Cracks

POINT DEFECTS

Point Defects are the irregularities or deviations from ideal arrangement around a point or an atom in a crystalline substance.



VACANCY DEFECT

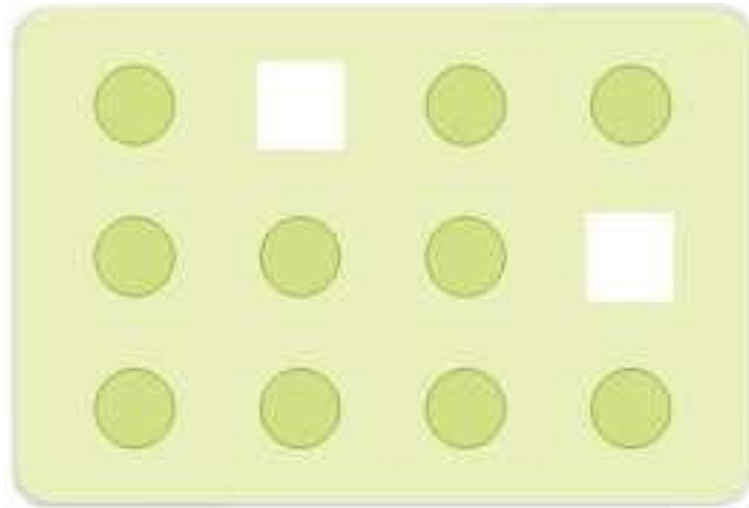
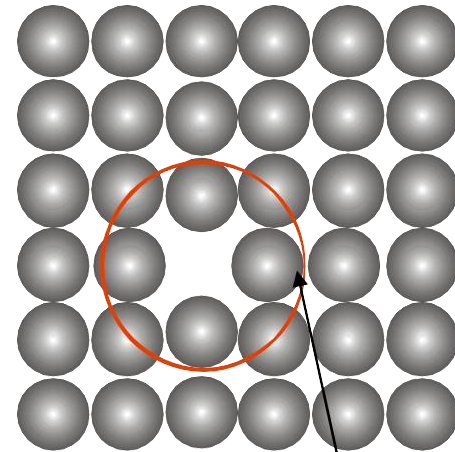


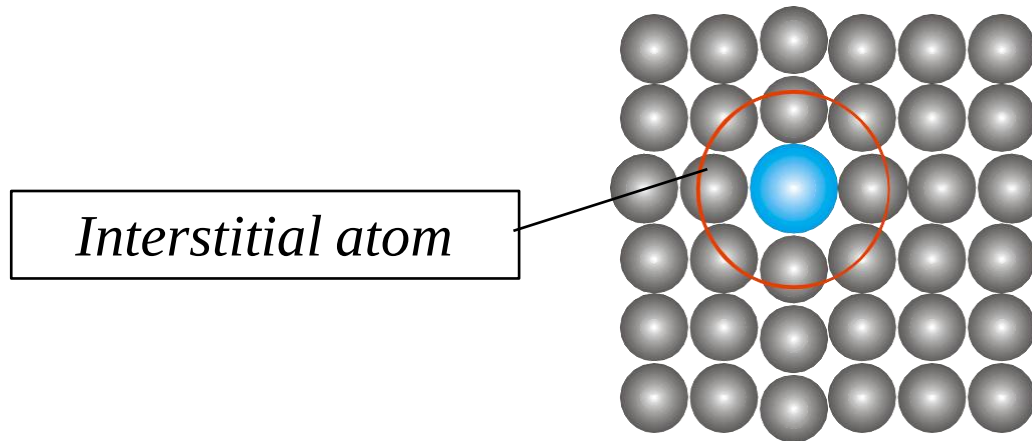
Fig. 1.23: Vacancy defects



Missing atom

- ☐ Atom missing from an atomic site
- ☐ Occur due to imperfect packing during crystallisation
- ☐ This results in decrease in density of the substance
- ☐ Number of vacancy defects depend on temperature

INTERSTITIALS DEFECT



- ☐ Addition of an extra atom within a crystal structure
- ☐ This defect increases the density of the substance
- ☐ Causes atomic distortion
- ☐ Vacancy and interstitials are inverse phenomena

SCHOTTKY DEFECT

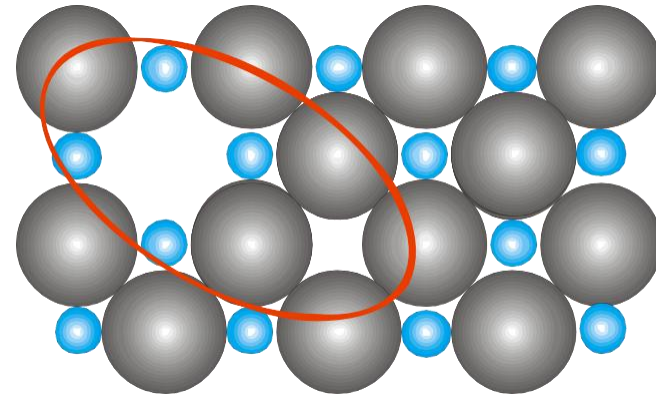
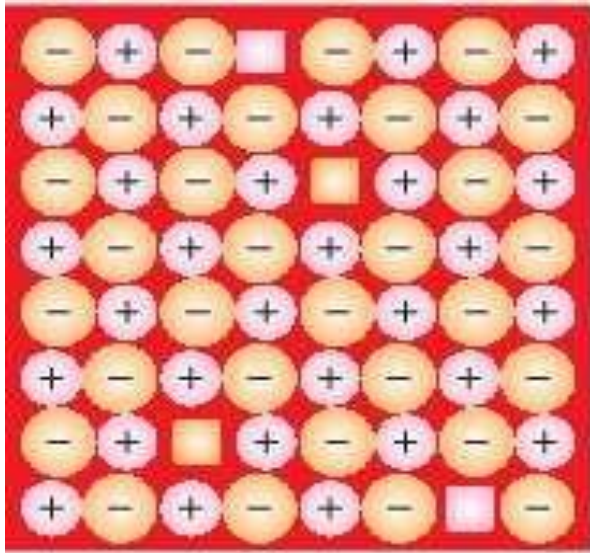


Fig. 1.26: Schottky defects

- ☐ Pair of anion and cation vacancies
- ☐ In order to maintain electrical neutrality, the number of missing cations and anions are equal
- ☐ It also decreases the density of crystal
- ☐ E.g. Alkali halides such as NaCl, KF, etc.

FRENKEL DEFECTS

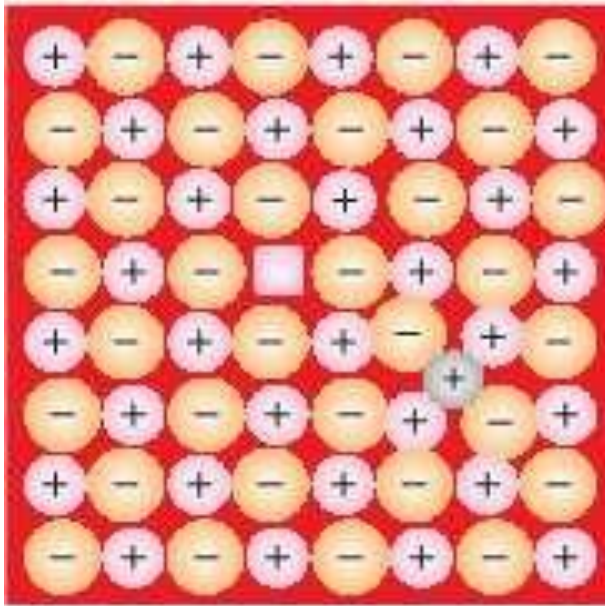
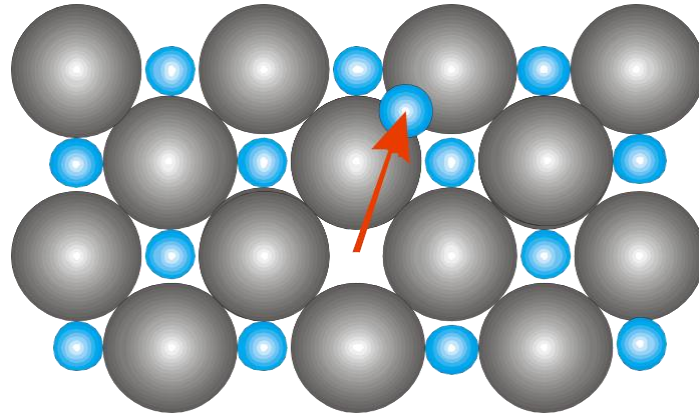


Fig. 1.25: Frenkel defects



- ☐ Cation (being smaller get displaced to interstitial voids)
- ☐ Combination of vacancy and interstitial atom
- ☐ No change in the density
- ☐ E.g. AgI, CaF_2